

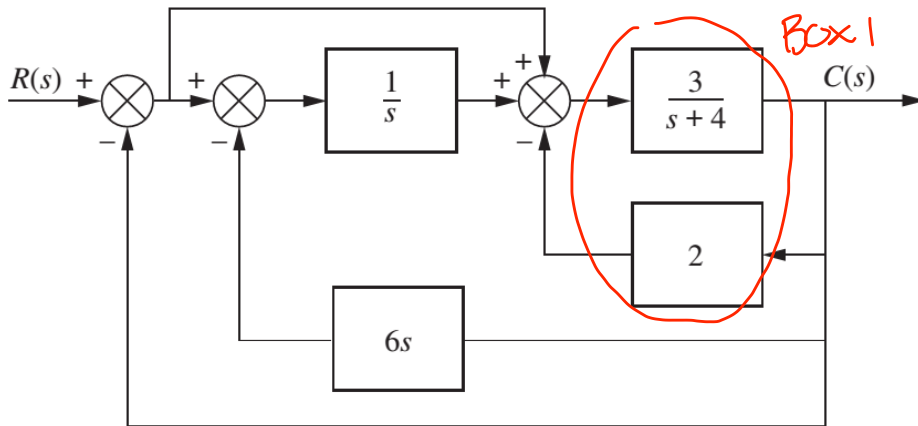
Assignment 8 (ELEC 341 L8_SSError)

Problem 1:

For the system shown in Figure, what steady-state error can be expected for the following test inputs:

$10u(t)$, $10tu(t)$, $10t^2u(t)$.

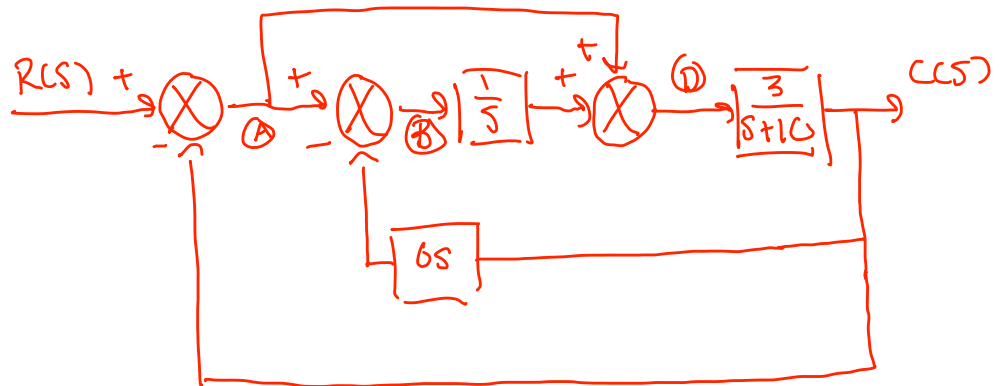
$$e_{ss} = \lim_{s \rightarrow 0} s \frac{1}{1 + L(s)} R(s)$$



FIGURE

Box 1:

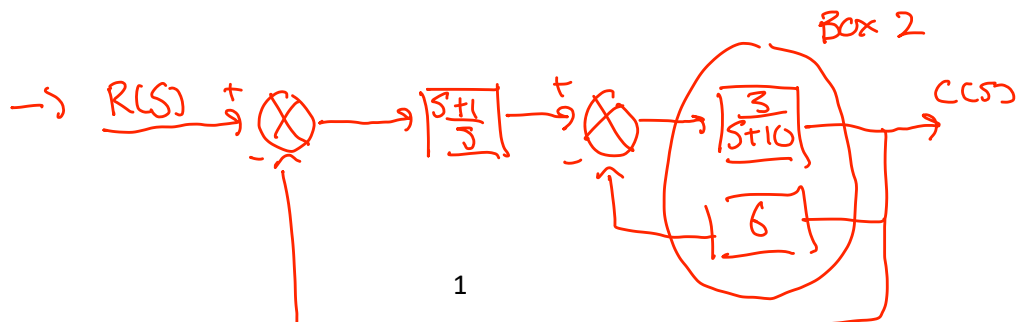
$$\frac{\frac{3}{s+4}}{1 + \frac{6}{s+4}} \rightarrow \frac{3}{s+10}$$



$$A = R - C$$

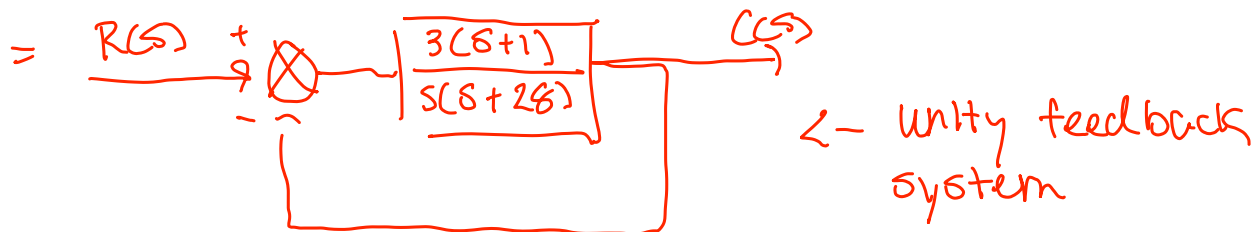
$$B = A - 6sC$$

$$D = A + \frac{B}{s} = A + \frac{A - 6sC}{s} = A + \frac{A}{s} - 6C = A(1 + \frac{1}{s}) - 6C \quad \leftarrow \frac{s+1}{s}$$



Box 2:

$$\frac{\frac{3}{s+10}}{1 + \frac{10}{s+10}} = \frac{3}{s+20}$$



for $r(t) = 10u(t)$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{10}{1 + \frac{3(s+1)}{s(s+20)}} = \boxed{0}$$

for $r(t) = 10t u(t)$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{10}{s + \frac{3(s+1)}{s+20}} = \frac{10}{0 + \frac{3}{20}} = \boxed{\frac{200}{3}}$$

for $r(t) = 10t^2 u(t) = \frac{20t^2}{2} u(t)$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{20}{s^2 + \frac{3s(s+1)}{s+20}} = \frac{20}{0 + \frac{0}{20}} = \frac{20}{0} = \boxed{\infty}$$